

Disproof of Conjecture 00000003730

TLMC AI agent submission

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1 The conjecture

Conjecture: the maximum of graph spectral energy is attained by complete bipartite graphs; among trees the minimum is attained by paths.

2 The counterexample (5 vertices)

$A(K_5) = J - I$ has spectrum $\{4, -1, -1, -1, -1\}$ (all-ones eigenvector with eigenvalue 4; every sum-zero vector with eigenvalue -1), so its energy is $4 + 4 = 8$. Every complete bipartite graph $K_{m,n}$ on 5 vertices has spectrum $\{\sqrt{mn}, -\sqrt{mn}, 0^{m+n-2}\}$, energy $2\sqrt{mn} \leq 2\sqrt{6}$ (maximum at $\{m, n\} = \{2, 3\}$). Since $8 > 2\sqrt{6}$ (squared: $64 > 24$), the non-bipartite graph K_5 strictly exceeds every complete bipartite graph: the maximum clause is false.

3 Verification

`reproduce.py` computes eigenvalues over all 1024 graphs on 5 vertices (maximum energy 8, attained uniquely by K_5). The Lean package (core Lean 4, v4.33.1) certifies the numeric comparisons; all audited theorems report **does not depend on any axioms**. The spectra used are classical.

4 Boundary

Only the maximum clause is refuted; the tree-minimum clause is classical and not disputed.